



Storm Shelter Challenge

My name _____

Class _____

Date _____



How do we build a shelter that stays standing and dry in a storm?

Big storms bring strong wind and heavy rain. People need shelters that stay standing and keep them dry. Can you build one?



My kit tick when you have it

- ☐ LEGO bricks and a base plate
- ☐ Card for a roof
- ☐ A small piece of tissue
- ☐ The same number of bricks for two shelters
- ☐ A desk fan with three speeds
- ☐ Card for a flat roof and a sloped roof
- ☐ A spray bottle of water
- ☐ Two dry pieces of tissue



Stay safe

- Keep small bricks away from little children.
- Keep fingers away from the fan blades.
- An adult switches the fan on and off.
- Keep water far away from the fan and plugs.
- Wipe up spills so nobody slips.

START HERE: BUILD



Build your shelter

1 · DO IT: TICK EACH STEP

- 1 ☐ Start on a base plate.
- 2 ☐ Build four walls. Leave a door.
- 3 ☐ Put a roof on top.
- 4 ☐ Put the tissue inside. It shows if rain gets in.

2 · LOOK & RECORD: CIRCLE WHAT YOU SEE

	Does it stand up by itself?
My shelter	Yes, strong Yes, but wobbly No, it falls

Draw your shelter

WORDS TO USE

walls roof door strong wobbly shelter bricks

TEST 1

🌬️ Tall or wide?

? Which shelter stands up to stronger wind?

1 · MY GUESS: CIRCLE ONE



Tall and thin



Low and wide



Both the same

How sure am I?



2 · DO IT: TICK EACH STEP

- 1 ☐ 🏠 Build a tall, thin shelter.
- 2 ☐ 🏠 Build a low, wide shelter with the same bricks.
- 3 ☐ 📏 Put the fan one ruler away from the shelter.
- 4 ☐ 🌿 Try low, then medium, then high.
- 5 ☐ 🖐️ Tap the speed that knocked it over.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	Which fan speed knocked it over?
🏠 Tall shelter	🌿 Low 🌬️ Medium 🌀 High 🖐️ It never fell
🏠 Wide shelter	🌿 Low 🌬️ Medium 🌀 High 🖐️ It never fell



Draw your shelter: Tall shelter



Draw your shelter: Wide shelter

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right
 ☐ 🌑 Partly right
 ☐ 🔄 Something different happened
 ☐ 🤔 Not sure yet

WORDS TO USE

wind fan tall wide fell strong blew stable

TEST 2

☁️ Flat roof or sloped roof?

? Which roof keeps the tissue drier?

1 · MY GUESS: CIRCLE ONE



A flat roof



A sloped roof



Both the same

How sure am I?



2 · DO IT: TICK EACH STEP

- 1 ☐ 🏠 Put a flat roof on. Put dry tissue inside.
- 2 ☐ 🧴 Spray the roof 10 times from above.
- 3 ☐ 👁️ Look at the tissue. Tap what you see.

- 4 ☐  Fold the card into a slope. Use new tissue.
- 5 ☐  Spray 10 times again, then look.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How is the tissue inside?
▬ Flat roof	 Dry  A bit damp  Soaked
▲ Sloped roof	 Dry  A bit damp  Soaked

 Draw your shelter: Flat roof

 Draw your shelter: Sloped roof

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right ☐ Partly right ☐ Something different happened ☐ Not sure yet

WORDS TO USE

rain roof flat slope dry damp soaked drip

Art corner: My storm-proof house

Draw and colour your dream storm-proof house.
Label the strong walls and the roof.

÷ Maths corner: Brick count

Count the bricks in one wall. How many walls? Draw them.

New words

Shelter: A place that keeps you safe from weather.

Engineer: Someone who designs and builds things to solve problems.

Stable: Hard to knock over.

Slope: A surface that goes up or down at an angle.

★ **What I can do now** colour the stars

I can build a shelter that stands up by itself.



I can test my shelter with wind and rain, fairly.



I can improve my design after testing it.

